Tutorato di algoritmi e laboratorio 04

Il tutor

Ma quanto è bello non trovate?

So che state trattando le equazioni di ricorrenza... Mi ricordate e spiegate le definizioni di $O, \Omega, \Theta, \omega$ e o?

O : CASO PESSIMO

$O(g(n))$

$f(n) \in O(g(n)) \iff$

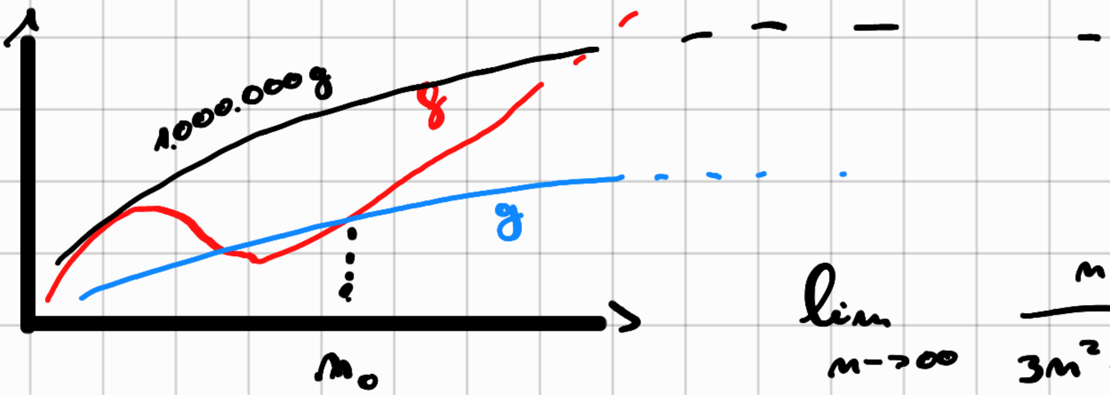
$\exists\, c \in \mathbb{R},\ n_0 \in \mathbb{N} : \forall n > n_0$

$0 < c f(n) \le g(n)$

$\exists\, c > 0,\ n_0 \in \mathbb{N} : \forall n > n_0$

$0 < f(n) \le c\, g(n)$ $\qquad$ $f(n) \ge c\, g(n)$

O $\qquad$ Ω



$$\lim_{n \to \infty} \frac{n^2}{3n^2 + 1 + \log n + 100n}$$

Mi spiegate cosa è un'equazione di ricorrenza?

$$F(n) = 1+2-3+F(n) = F(n)$$

$$\begin{cases} F(n+2) = F(n+1) + F(n) \\ F(0) = F(1) = 1 \end{cases}$$

$$\begin{cases} F(n) = F(n-2) + 1 \\ F(0) = 1 \end{cases}$$

$$F(3) = F(1) + 1 = F(-1) + 1 + 1 = F(-3) + 1 + 1 + 1 + 1$$

Esempi:

- Fattoriale
- Fibonacci
- Massimo di un albero binario bilanciato
- Mergesort

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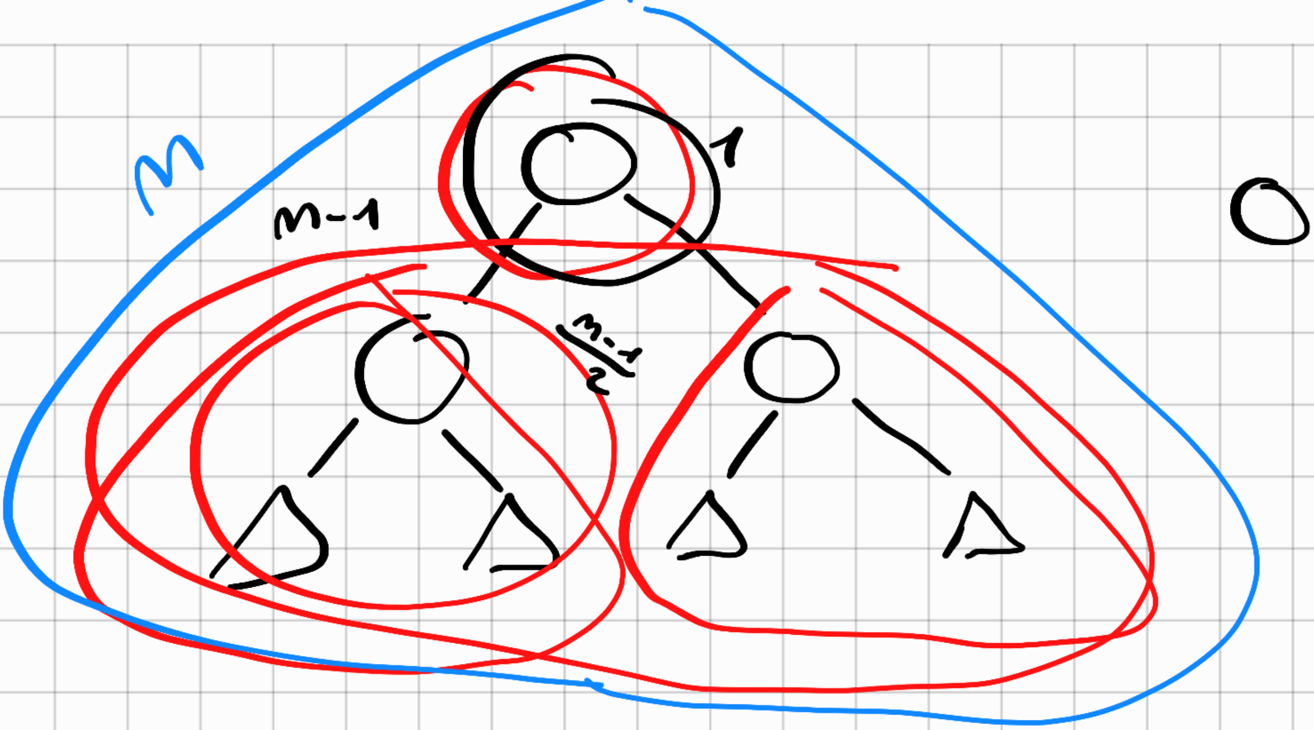
Fact(n)
  IF (n = 0)              1
    RETURN 1              0
  RETURN n · FACT(n-1)    T(n-1) + 2
```

$T(n) = T(n-1) + 3$

$T(0) = 1$



max-Key (T) $T(2^{h-1})$

if (T è fatto solo dalla radice)
  RETURN Key(root)
$\Big\}\ \Theta(1)$

max ← Key(root) $\Theta(1)$

a ← max-Key(left(T)) $T\left(\frac{n-1}{2}\right) = T(2^{h-1}-1)$

b ← max-Key(right(T)) $T\left(\frac{n-1}{2}\right)$

if (a > max) then
  max ← a
$\Big\}\ \Theta(1)$

if (b > max) then
  max ← b
$\Big\}\ \Theta 1$

RETURN max

$T(n) = 2T\left(\frac{n-1}{2}\right) + \Theta(1)$

Merge-sort(A, n)

  If $(n=1)$

    return A $\Big\}$ $\Theta(1)$

  $B \leftarrow$ Merge-sort$(A, \lfloor \frac{n}{2} \rfloor)$ $T(\lfloor \frac{n}{2} \rfloor)$

  $C \leftarrow$ Merge-sort$(A_{\frac{n}{2}}, \lceil \frac{n}{2} \rceil)$ $T(\lceil \frac{n}{2} \rceil)$

  $A \leftarrow$ Merge(B, C) $O(n)$

  return A

```
Insertion-sort (A, n)
  if (n = 1)
    return A
  Insertion sort (A, n-1)
  insert (A, A[n-1])
return A
```

$\left.\begin{array}{l}\text{if } (n=1)\\ \quad\text{return } A\end{array}\right\}\ \Theta(1)$

Insertion sort (A, n-1) $T(n-1)$

insert (A, A[n-1]) $\Theta(n)$

Esercizio che farete voi!

Scrivere lo pseudocodice di:

- Heap-Sort
- Heapify
- Counting-sort
- Radix-sort

Prima di salutarci...

Belfun (p, s, n)

if $(n == 0)$ then

$\left.\begin{array}{l} \\ \text{return } p \cdot s \end{array}\right\} \Theta(1)$ concatenazione stringhe

return $\underbrace{\text{Belfun}(p, s, \lfloor \frac{n}{2} \rfloor)}_{T(\lfloor \frac{n}{2} \rfloor)} \cdot \underbrace{\text{Belfun}(\text{""}, s, \lfloor \frac{n}{2} \rfloor)}_{T(\lfloor \frac{n}{2} \rfloor)}$

$T(n) = ?$ $\qquad T(\lfloor \frac{n}{2} \rfloor) + T(\lfloor \frac{n}{2} \rfloor) = 2T(\lfloor \frac{n}{2} \rfloor)$

In coro

Ditemi dunque... Belfun("b", "c", 4) = ?